Math 54 Discussion Worksheet

Name: ______________________________ ID: ____________________

Problem 1

Use Cramer's Rule to solve the following system of equations:

$$\begin{aligned} 4x_1 + x_2 &= 6 \\ 3x_1 + 2x_2 &= 5 \end{aligned}$$

Problem 2

Compute the inverse of the following matrix using Cramer's rule, i.e. compute the adjugate and the determinant.

$$\begin{bmatrix} 5 & 0 & 0 \\ -1 & 1 & 0 \\ -2 & 3 & -1 \end{bmatrix}$$

Problem 3

Let J be the $n \times n$ matrix of all 1's, and consider the $n \times n$ matrix

$$A = (a-b)I + bJ,$$

that is,

$$A = \begin{pmatrix} a & b & b & \cdots & b \\ b & a & b & \cdots & b \\ b & b & a & \cdots & b \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ b & b & b & \cdots & a \end{pmatrix}.$$

Confirm that

$$\det(A) = (a-b)^{n-1}\big(a + (n-1)b\big)$$

as follows:

(a) Subtract row 2 from row 1, row 3 from row 2, and so on. Explain why this does not change the determinant.

(b) With the resulting matrix from part (a), add column 1 to column 2, then add this new column 2 to column 3, and so on. Explain why this does not change the determinant.

(c) Find the determinant of the resulting matrix from part (b).

Now define

$$B = \begin{pmatrix} a-b & b & b & \cdots & b \\ 0 & a & b & \cdots & b \\ 0 & b & a & \cdots & b \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & b & b & \cdots & a \end{pmatrix}, \qquad C = \begin{pmatrix} b & b & b & \cdots & b \\ b & a & b & \cdots & b \\ b & b & a & \cdots & b \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ b & b & b & \cdots & a \end{pmatrix}.$$

Notice that the first column of A equals the sum of the first columns of B and C. Using the linearity property of the determinant in a single column, show that

$$\det(A) = \det(B) + \det(C),$$

and use this to prove the formula for $\det(A)$ above by induction on n.